

Single Inheritance in Java

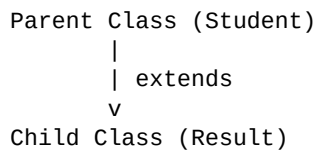
Java Programming Lab Record

1. Aim

To implement Single Inheritance in Java, execute the program using multiple test cases, and verify the results.

2. Concept

Single inheritance occurs when one child class inherits the properties and methods of one parent class using the extends keyword.



3. Java Program

```
import java.util.Scanner;

// Parent class
class Student {
    String name;
    int marks;

    void getStudentDetails(String name, int marks) {
        this.name = name;
        this.marks = marks;
    }
}

// Child class
class Result extends Student {

    void displayResult() {
        System.out.println("Student Name: " + name);
        System.out.println("Marks: " + marks);

        if (marks >= 35) {
            System.out.println("Result: Pass");
        } else {
            System.out.println("Result: Fail");
        }
    }
}

// Main class
public class SingleInheritance {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of test cases: ");
        int n = sc.nextInt();
        sc.nextLine();

        for (int i = 1; i <= n; i++) {

            System.out.println("\n----- Test Case " + i + " -----");
```

```

        System.out.print("Enter student name: ");
        String name = sc.nextLine();

        System.out.print("Enter marks: ");
        int marks = sc.nextInt();
        sc.nextLine();

        // Creating child class object
        Result r = new Result();

        // Calling inherited method
        r.getStudentDetails(name, marks);

        // Calling child class method
        r.displayResult();
    }

    sc.close();
}

```

4. Explanation

- **Student** is the parent/superclass.
- **Result** is the child/subclass.
- **Result extends Student** establishes inheritance.
- The Result object can access `getStudentDetails()` because it is inherited from Student.
- `displayResult()` belongs to the Result class.
- If marks are 35 or above, the result is Pass; otherwise, it is Fail.
- The program takes the number of test cases as input, then loops that many times, reading a fresh name and marks and creating a new Result object on each iteration.

5. Test Cases

Test Case	Student Name	Marks	Expected Result	Actual Result
1	dinkar	34	Fail	Fail ■
2	guna	40	Pass	Pass ■
3	ravan	78	Pass	Pass ■

6. Output Screenshot

The screenshot below shows the program actually executed in Eclipse, running all three test cases in a single session:

```
<terminated> Singleinheritance [Java Application] C:\Users\togid\Down
Enter number of test cases: 3

----- Test Case 1 -----
Enter student name: dinkar
Enter marks: 34
Student Name: dinkar
Marks: 34
Result: Fail

----- Test Case 2 -----
Enter student name: guna
Enter marks: 40
Student Name: guna
Marks: 40
Result: Pass

----- Test Case 3 -----
Enter student name: ravan
Enter marks: 78
Student Name: ravan
Marks: 78
Result: Pass
```

The results verify that the program works correctly for both passing and failing marks, confirming that single inheritance via extends functions as expected.